

C / C++ Programming Syllabus



a1training167@gmail.com



www.a1training.in



8368979712



6380486914



C-167, Omicron 1, 6% Abadi, Greater Noida / Earthcon Sanakriti, Noida Extension



SYLLABUS FOR C & C++

Programming in 'C'

CHAPTER - 1 Introduction of Programming Languages

1.1 Types of Languages

1.2 Evolution of 'C' Language

1.3 Structure of a 'C' Program

1.4 'C' Program development life cycle

1.5 Executing and Debugging a 'C' Program

CHAPTER – 2 'C' Tokens

2.1 Keywords and Identifiers

2.2 Operators

2.3 Constants

2.4 Variables

2.5 Data Types

2.6 Precedence of Operators

2.7 Scope and Lifetime of Variables





CHAPTER – 3 Control Statement and Expressions

3.1 Decision Making using if statement

3.2 Types of if ...else block

3.3 Switch case Block

3.4 Arithmetic Expressions

3.5 Evaluation of Expressions

3.6 GOTO statement

CHAPTER – 4 Looping

4.1 Concept of Loop

4.2 For loop 4.3 While loop

4.4 Do while loop

4.5 Jumping in Loop

4.6 break and continue statement

CHAPTER – 5 Arrays and String

5.1 Introduction of Array





5.2 One - D Array

5.3 Two - D Array

5.4 Multidimensional Array

5.5 Dynamic Arrays

5.6 Implementing String Variables

5.7 String handling Functions

CHAPTER – 6 Functions

6.1 Concept of Function

6.2 User defined Function

6.3 System Defined Function

6.4 Types of parameter passing in function

CHAPTER – 7 Pointers

7.1 Need of Pointers

7.2 Types of Pointers

7.3 Pointer Expression

7.4 Arrays of Pointers



7.5 Pointers and Functions

CHAPTER - 8 Structure and Unions

8.1 Need of Structure

8.2 Implementing Structure Variable

8.3 Arrays of Structure

8.4 Structure within Structure

8.5 Introduction of Unions

8.6 Difference between Structure and Unions

CHAPTER – 9 File Handling using 'C'

9.1 Opening and Closing File

9.2 Input / Output operations on File

9.3 Random Access to Files

9.4 Command Line Arguments

CHAPTER - 10 Dynamic Memory Allocation

10.1 Concept of Dynamic Allocation





10.2 Implementing Malloc and Calloc Functions

10.3 Releasing the free space

CHAPTER - 11 Storage Classes and Preprocessor

11.1 Introduction of Storage Class

11.2 Types of Storage Classes

11.3 Introduction of Pre-processor

11.4 Macro Substitution

11.5 File Inclusion Programming in 'C++'

CHAPTER - 12 Introduction to Object Oriented Programming

12.1 Concept of OOP

12.2 Features of OOP

12.3 Introduction of 'C++'

12.4 Structure of 'C++' program

12.5 Executing and Debugging a 'C++' Program





CHAPTER - 13 'C++' Tokens and Type Casting

13.1 Keywords and Identifiers

13.2 Operators

13.3 Constants

13.4 Variables

13.5 Data Types

13.6 Precedence of Operators

13.7 Scope and Lifetime of Variables

CHAPTER – 14 Classes & Objects

14.1 Classes & Object Specifier

14.2 Defining data members and member functions

14.3 Array of objects

14.4 Managing console I/O

14.5 'C++' stream classes

14.6 Formatted and unformatted console I/O

14.7 Usage of manipulators





CHAPTER – 15 Function in ‘C++’

15.1 Call by reference, Return by reference

15.2 Function overloading and default arguments

15.3 Inline function

15.4 Static class members

15.5 Friend functions

15.6 Virtual Functions

CHAPTER - 16 Constructors and Destructor

16.1 Concept of Constructor

16.2 Types of Constructors

16.3 Memory allocation (new and delete)

16.4 Usage of destructor

CHAPTER - 17 Operator Overloading

17.1 Overloading Unary and Binary operators

17.2 Overloading using friend function





CHAPTER – 18 Inheritance

18.1 Types of inheritance

18.2 Virtual base classes and abstract base classes

18.3 Constructor and destructor in derived class

CHAPTER - 19 Working with files

19.1 File operations

19.2 File pointer and their manipulation

19.3 File updation with random access

CHAPTER – 20 Exception Handling

20.1 Various Exception Handling classes

20.2 Implementing try and catch block

20.3 Use of throw keyword

